Task 1

Use the internet and textbooks to research four of the scientists involved in the discovery of the DNA model. Complete the fact file for each scientist.

James Watson

Insert a picture of James Watson

Born:

Died:

Education:

Scientific work:

Francis Crick

Insert a picture of Francis Crick

Born:

Died:

Education:

Scientific work:

Maurice Wilkins

Insert a picture of Maurice Wilkins

Born:

Died:

Education:

Scientific work:

Rosalind Franklin

Insert a picture of Rosalind Franklin

Born:

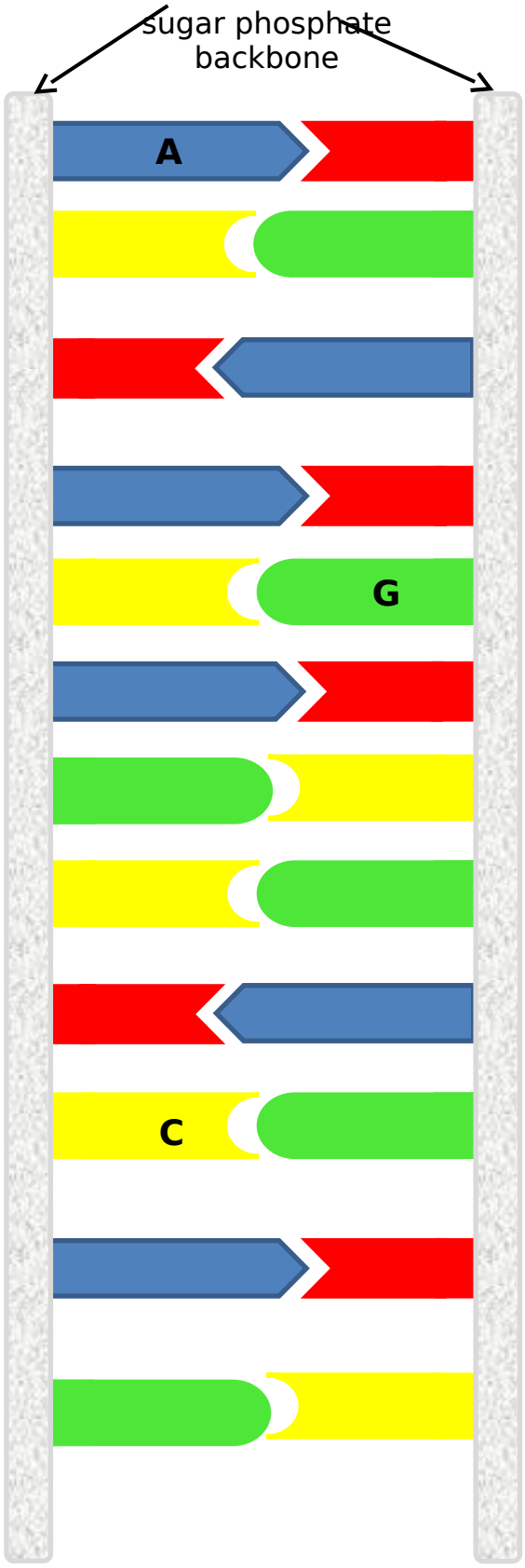
Died:

Education:

Scientific work:

Task 2 – can you crack the code?

Use your powers of deduction (and research skills) to crack the DNA code and label the diagram.



There are 4 DNA bases. They are represented the letters:

A C G T

1) Find out what these letters stand for:

A =

C =

G =

T =

The bases always pair up in a specific way.

2) Some of the bases have been labelled on the diagram. Use these to work out which bases always pair together.

____ always pairs with ____

and

____ always pairs with ____

3) Complete the diagram by labelling all the bases.

Teaching notes and suggested answers

The following websites are a useful place to start to research the scientists (links available at the time of publication).
http://www.famousscientists.org/
http://www.bbc.co.uk/science/0/22270604
http://www.dnai.org/timeline/

http://www.chemheritage.org/discover/online-resources/chemistry-in-history/themes/biomolecules/dna/watson-crick-wilkins-franklin.aspx

Task 1

James Watson

Born: 1928, Chicago, Illinois.
Died: Still alive
Education: Attended University of Chicago, Indiana University and the Cavendish Lab of the University of Cambridge
Scientific work:

- 1953 - Watson and Crick, using data collected by Wilkins and Franklin, discovered the double helix structure of DNA.
- Watson and Crick submitted a paper to the scientific journal *Nature*, published on April 25, 1953.
- 1962 – with Crick and Wilkins he won the Nobel Prize in physiology.

Francis Crick

Born: 1916, Northampton, England
Died: 2004
Education: Northampton Grammar School, Mill Hill School London, University College London, University of Cambridge (PhD)
Scientific work:

- 1953 - Watson and Crick, using data collected by Wilkins and Franklin, discovered the double helix structure of DNA.
- Watson and Crick submitted a paper to the scientific journal *Nature*, published on April 25, 1953.
- 1962 – with Crick and Wilkins he won the Nobel Prize in physiology.
- Predicted the mechanism for protein synthesis.

Maurice Wilkins

Born: 1916 Pongaroa, New Zealand
Died: 2004
Education: King Edward's School, Birmingham and St John's College, Cambridge
Scientific work:

- He worked as part of a research team with Rosalind Franklin at King's College, London.
- They used x-ray crystallography to study DNA structure.
- Nobel Prize winner in 1958 with Watson and Crick.

Rosalind Franklin

Born: 1920, London
Died: 1958
Education: St Paul's Girls' School, London, Newnham College Cambridge
Scientific work:

- Worked for the Coal Utilization Research Association on the microstructure of coke using gas chromatography.
- Set up a lab to study DNA using x-ray crystallography.
- Also studied the structure of plant viruses.

Task 2

1) A = **adenine**, T = **thymine**, C = **cytosine**, G = **guanine**

2) A always pairs with **T** and **C** always pairs with **G**.

3)

